Standard operating procedures for field implementation and data collection of on-farm trials with the tricot approach: sorghum

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Crop: sorghum

Scientific name: *Sorghum bicolor* (L.) Moench

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Traits: Germination, Diseases resistance, Drought tolerance, Striga resistance, Antracnosis resistance, Plant survival, Panicle length, Maturity, Threshabilty, Stover yield, Yield, Grain size, Grain hardness, Grain colour, Malting quality, Taste, Marketability, Overall preference, Overall vs Local

Data collection moments: Participant registration form, Emergence, Vegetative, Physiological maturity, Technician assessment - Reproductive, Farmer assessment - Harvest, Technician assessment - Harvest, Socio-economic baseline, Data manager inputs

This document is also available in: PT, FR, ES, SWH

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Introduction

This document provides a comprehensive overview of the data to be collected during a sorghum on-farm trial implemented using the tricot approach and designed on ClimMob platform (https://climmob.net). The protocol outlines the key data collection points throughout the trials, including the variables and traits to be collected at each crop growth stage. Farmers evaluate varieties in their on-farm trials and provide comparative observations by ranking the varieties based on their relative performance throughout their growth and post-harvest qualities. A more in-depth overview of the steps required for developing a participatory on-farm trial using tricot is available here https://learn.climmob.net.

Scope

Coverage: This SOP describes the procedures and standards for implementing decentralized participatory on-farm trials using the tricot approach. It covers the evaluation design, documentation, monitoring, and data management of tricot trials. The SOP includes guidance on digital survey configuration, data collection, metadata standards, and basic quality assurance procedures to support FAIR data generation. This SOP is intended to support standardized implementation of tricot trials while allowing contextual adaptation to local crops, farming systems, languages, and institutional settings.

Exclusions: This SOP does not cover guidance on selection of genotypes, selection of participants, advanced statistical analysis, breeding decision-making workflows, laboratory analysis procedures, seed multiplication and certification protocols. Specific procedures related to genomic analysis, climate modelling, seed shipment regulations, or laboratory phenotyping should be addressed in separate SOPs where applicable.

Intended audience: This SOP is intended for breeders, agronomists, genebank staff, NGOs, farmer organizations, students, and researchers involved in the implementation, coordination, and management of tricot trials and decentralized participatory experimentation.

Informed consent

The informed consent form provides a standardized framework to ensure that all participants are adequately informed before taking part in the experiment and associated surveys. It explains the purpose of the data collection, the types of information that will be collected, and how these data may be used, including for research and publication. Participation is entirely voluntary. Participants may decline to answer any question or withdraw from the study at any time, without any negative consequences.

Consent is obtained before data collection to ensure transparency, ethical research practice, and respect for participants' rights. Providing informed consent is a prerequisite for inclusion in all subsequent trial and survey activities. Informed consent template is appended to this SOP.

Trial design

All plots well are labelled using letters (sub-plot A, sub-plot B, and sub-plot C). Sub-plots should be established on the same planting date and follow the same dimensions and management practices. The three sub-plots should be located very close to ensure exposure to similar environmental and management conditions.

If sufficient land is available, farmers are encouraged to establish an additional sub-plot of their preferred or commonly grown local variety ("Local") using the same planting date, dimensions,

and management practices. Although the Local sub-plot is not part of the core tricot design, it is included in the end-of-season evaluation, where participants are asked whether the local variety performed better or worse than options A, B, and C.

Sub-plot size should follow recommendations from the Product Management Team (PMT) and be appropriate for the crop and farming system. Information on sub-plot dimensions should be recorded in the trial metadata. For sorghum the advised sub-plot size is 15 m² (4 rows, 5 m row length, 75 cm between rows and 30 cm between plants), which requires 400 seeds (10 g of seed per variety) per sub-plot.



Materials and equipment

Data collection in tricot trials may be conducted using either digital surveys administered by field staff or observation cards completed by participants. Depending on the selected data collection approach, the following materials and equipment may be required.

Digital data collection (ODK): i) Android smartphone or tablet with sufficient battery capacity; ii) ODK Collect application installed and configured; iii) Internet connection for downloading and submitting forms (when available); iv) Power bank or charging equipment for fieldwork; v) Participant lists and trial allocation records.

Observation card-based data collection: i) Printed trial observation cards; ii) Pens or pencils for participants; iii) Trial labels identifying options A, B, and C; iv) Participant lists and trial allocation records.

Data entry and compilation: i) Computer with internet access; ii) Access to the ClimMob platform; iii) Enketo web forms for direct data entry or iv) ClimMob Excel templates for offline data entry and subsequent upload.

Project teams should select the data collection method most appropriate for local conditions, participant preferences, and available resources.

Trial observation and data collection

Participants observe their assigned plots throughout the trial duration and record their observations at predefined data collection moments (crop growth stages). Data may be

collected directly by field technicians using ODK on Android devices or recorded by participants on trial observation cards and subsequently digitized in ClimMob using Enketo forms or Excel templates. This SOP defines the observation moments, which are aligned with crop development stages and should be clearly communicated to participants during the distribution of the trial package and throughout the trial period.

Observation cards should contain simple comparison questions that allow participants to identify the best and worst option for each evaluation criterion. Participants should be encouraged to focus on one criterion at a time when making their assessments. Project teams are encouraged to provide follow-up support through field visits, phone calls, or other communication channels to remind participants of upcoming observations, answer questions, and support timely data collection.



Participant registration

Participant registration is conducted during the distribution of trial packages and should be completed for all participants before any subsequent data collection activities can take place. Trial packages should be distributed sufficiently in advance of planting to allow participants adequate time for land preparation and trial establishment. The purpose of registration is to link each participant to a unique trial package and collect the minimum information required for trial management and participant follow-up.

Registration activities should remain as brief and efficient as possible, as package distribution is often a busy activity during which project staff must distribute trial materials, explain trial procedures, answer participant questions, and provide implementation guidance. Completion of participant registration is mandatory for inclusion in subsequent data collection activities and analyses. During registration, project staff should explain the trial objectives, plot establishment procedures, observation schedule, and participant responsibilities throughout the duration of the trial.

Socio-economic baseline survey

The socio-economic baseline survey collects information on participant characteristics, household composition, farming activities, crop management practices, seed systems, market

participation, and access to agricultural services. The purpose of this survey is to identify socio-economic factors associated with crop variety management and farmer preferences, and to support the interpretation of trial results across different participant groups.

Because the socio-economic survey is more comprehensive than the participant registration form, it does not need to be administered at the same time as trial package distribution. The survey may be conducted during follow-up visits, or at any suitable moment during the trial period. Project teams may administer the survey to different groups of participants throughout the duration of the trial according to available resources and operational constraints. Although completion of the socio-economic survey is encouraged for all participants, project teams may prioritize a representative subset of participants when resources are limited. The survey should be completed before final data analysis whenever possible to enable the integration of socio-economic information with trial observations and participant evaluations.

Survey components

The tricot data collection framework is composed of multiple survey components implemented throughout the duration of the trial. These components collect complementary information required for trial management, participant follow-up, characterization of trial conditions, and evaluation of technology performance. Data collection begins with participant registration and informed consent, followed by crop-stage observations recorded throughout the trial. Depending on project objectives and available resources, additional socio-economic information may be collected from all participants or from a representative subset of participants.

At the end of the trial, participants complete final evaluations of the tested technologies, including post-harvest observations. The following sections provide the complete list of variables, questions, and data collection moments associated with this tricot protocol. This survey is available on ClimMob as a project template named Sweetpotato OFVT Breeding.

Participant registration form

Question name	Question description
Consent	Are you willingly to accept hosting a tricot trial and provide required information from the trial?
Participant name	Participant name
Point of delivery	Please register the GPS location where the seeds were delivered
ADM1	Name of administrative zone level 1 (state, province, region, etc)
ADM2	Name of administrative zone level 2 (city, county, town, local govern area, etc)
ADM3	Name of administrative region level 3 (community, village etc)
Age	Please indicate your age
Gender	Please select your gender
Experience with the crop	Please indicate how many years you have growing the crop?
Experience with tricot	Have you contributed to a tricot trial before? If yes, how many times? Zero if first time.

Emergence

Question name	Question description
On time	Was your trial package delivered on time?
Protocol	Have you followed the trial protocol (plot size, density) when establishing your

Question name	Question description
	trial?
Planting date	Date of trial establishment
Trial GPS	The GPS location of the on-farm trial
Previous land use	What was the previous land use of the on-farm plot site?
Mode of land preparation	What equipment was used for land preparation?
Slope	What is the slope of the area you established your tricot trial?
Plot size	What was the plot size?
Plot size unit	What is the unit used the measure the plot size
Local variety	What is the name of the local variety planted in the trial as check?
Germination	Which option germinated more plants?; Which option germinated less plants?
Diseases resistance	Which option is more resistant to diseases?; Which option is more susceptible to diseases?

Vegetative

Question name	Question description
Plant damage	Did any plant suffer damage during the trial?
Plant damage how	If any damage, do you know what was the cause for damage in
Diseases any	Did you find any disease in the trial?
Diseases where	Which part of the plant was affected by the disease?
Drought tolerance	Which option is more tolerant to drought?; Which option is more susceptible to drought?
Striga resistance	Which option is more resistant to striga?; Which option is more susceptible to striga?
Antracnosis resistance	Which option is more resistant to antracnosis?; Which option is more susceptible to antracnosis?
Plant survival	Which option has the higher plant survival rate?; Which option has the lower plant survival rate?

Physiological maturity

Question name	Question description
Panicle length	Which option has the larger panicles?; Which option has the smaller panicles?
Maturity	Which option has the earlier maturity?; Which option has the later maturity?

Technician assessment - Reproductive

Question name	Question description
Flowering (days to 50%)	Days to 50% flowering
Plant height	What is plant height of
Plant population	What is the number of plants in

Farmer assessment - Harvest

Question name	Question description
Irrigation	Did you use an irrigation system on your tricot trial?

Question name	Question description
Fertilizer	Did you use fertilizers on your tricot trial?
Fertilizer frequency	If yes, how many times you applied the fertilizer? Zero if none.
Pesticide	Did you use pesticides on your tricot trial?
Pesticide frequency	If yes, how many times you applied the pesticide? Zero if none.
Harvesting date	On which day did you harvest the plot
Threshability	Which option is easier to thresh? ; Which option is more difficult to thresh?
Stover yield	Which option has the higher stover yield?; Which option has the lower stover yield?
Yield	Which option has the higher yield?; Which option has the lower yield?
Grain size	Which option has the best grain/seed size for the target market?; Which option has the worst grain/seed size for the target market?
Grain hardness	Which option has the best hardness for the target use?; Which option has the worst hardness for the target use?
Grain colour	Which option has the best grain/seed colour for the target market?; Which option has the worst grain/seed colour for the target market?
Malting quality	Which option has better malting quality?; Which option has the worse malting quality?
Taste	Which option has the best taste?; Which option has the worst taste?
Marketability	Which option is easier to sell?; Which option is more difficult to sell?
Overall preference	Overall, which option was the best?; Overall, which option was the worst?
Best why	Why do you indicate this option as the best?
Worst why	Why you indicated this option as the worst?
Overall vs Local	Overall, is A, B, C better or worse than the one that you currently grow?
Grow again	If seeds were available for all three options, which option(s) would you like to grow in the next season?

Technician assessment - Harvest

Question name	Question description
Yield (quantitative)	The yield of
Yield unit	Which unit did you use to measure the yield?

Socio-economic baseline

Question name	Question description
Civil status	Please indicate your civil status
Occupation	What is the main occupation of the respondent?
Education	Please select your highest degree obtained
Education years	For how many years did you attend school?
Reading	Is the respondent able to read in English?

Question name	Question description
Reading Local Language	Is the respondent able to read in the local language?
Type of household	What is the type of household?
Household size	Number of persons living together and sharing meals for more than 6 months a year
Main crop	What is the most important crop that you produce?
Other crops	What are the crops that you produce?
Who controls seed exchange	Who in your household is mainly responsible for {your tricot crop} seed saving or seed exchange?
Varieties source	What is the source of {your tricot crop} seed?
Number of varieties	How many different varieties of {your tricot crop} does your household produce?
Variety name	What is the name of the variety(ies) that you commonly plant in your field?
Variety brand	Does the seed that you buy have a brand and a label? If yes, please indicate the name.
Ways of selling	How do you usually sell {your tricot crop}?
Who controls selling	Who in your household is mainly responsible for {your tricot crop} selling?
Distance to market	How far is the nearest market or place you sell your {your tricot crop} from your farm?
Sell share	What share of your {your tricot crop} harvest do you usually sell?
Income crop sell	What share of your total household income comes from selling {your tricot crop}?
Who controls production	Who in your household is mainly responsible for {your tricot crop} production?
Production last 12 months	Did you produced {your tricot crop} during the past 12 months?
Frequency of production	How many times per year you usually produce your {tricot crop}?
Reason no production	If no production during the past 12 months, why?
Production practices	Which of the following practices do you usually apply in {your tricot crop} production?
Member of an organization	To what types of organization do you belong?
Public extension service	Are you in contact with government/public extension services in relation with {your tricot crop} production?
Private extension service	Are you in contact with private extension services in relation with {your tricot crop} production?

Data manager inputs

Question name	Question description
Agroecological zone	Please indicate the agroecological zone where the plot was established

Other resources and SOPs

The following table provide additional guidance and SOPs for planning, implementing, documenting, and analysing tricot trials.

Resource	Link
Check list for implementing a tricot trial	https://learn.climmob.net/preparation/checklist
Budget for a tricot trial	https://learn.climmob.net/preparation/budgeting
Selection of participants	https://learn.climmob.net/design/sampling-plots
Selection of locations	https://learn.climmob.net/design/trial_size
Selection of genotypes	TBD
Seed multiplication	TBD
Providing metadata for your trial	https://learn.climmob.net/design/trial_documentation
Data analysis	https://learn.climmob.net/analysis/intro-data-analysis
Yield estimation	TBD